THANH VU

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RESEARCH INTERESTS

Model evaluation and probabilistic reasoning for messy real-world data; data systems that make ML reproducible.

EDUCATION

Franklin & Marshall College, Lancaster, PA *May 2026*

B.A. in Computer Science, Magna Cum Laude | GPA 3.72/4.0

- Honors List: Fall 2021, Fall 2023, Fall 2024, Spring 2025, Fall 2025; Dean's List: Spring 2022
- Computer Science: Machine Learning, Artificial Intelligence, Information Theory, Algorithms, Theory of Computer Science, Computer Organization
- Math & Statistics: Probability & Statistics, Statistical Modeling, Applied Regression, Linear Algebra & Differential Equations

RESEARCH & APPLIED PROJECTS

Can Mail Data Predict Housing Vacancy? | Microsoft Fabric, Power BI *Sep 2026*

- Built a reproducible Fabric pipeline testing whether quarterly USPS vacancy counts can proxy the annual ACS rate (6 counties, 84 county-years).
- Found pooled r = 0.77 but county-level r from −0.01 to 0.72 and 57.7% directional agreement: the pooled figure overstated the fit.
- Isolated a 2012 USPS classification change as a data artifact; fixed two silent ingestion bugs, preventing a false trend.

Decision Trees: co-written paper and AI-run follow-up | Information Theory, Python *Fall 2024; Oct 2026*

- Co-wrote an 18-page Information Theory course paper on entropy-based splits and pruning; in Oct 2026 an AI assistant designed, ran and wrote most of a follow-up testing its claims on 48,842 census records (disclosed in repo).
- Bias claim held: a 1,000-value noise feature was chosen just below the root in every run, cutting test accuracy 4.2–5.2 points; gain ratio only reduced it.
- Pruning claims did not: CV pre-/post-pruning lifted accuracy 81.8%→85.6%/86.0% with 19–50× fewer leaves, but widened the women–men false-negative gap (missed >$50K earners) from 6.7 to 9.9 points in 7 of 10 splits.

Machine Learning Coursework (CPS 360) | Python, NumPy *Spring 2026; AI review Oct 2026*

- Wrote k-means and linear/logistic gradient descent from scratch; an AI assistant's review later traced 90.7% accuracy (all false negatives) to an unconverged bias term; its standardizing fix raised it to 99.3% (scikit-learn: 99.5%).

Grocery Transaction Data Warehouse | SQL Server, T-SQL *Jul–Aug 2025*

- Built a Medallion warehouse over 131M+ transaction lines with stored-procedure ETL, a star-schema Gold layer, and 21 data-quality checks; tuning cut processing time 25%.

EXPERIENCE

Data Analyst Fellow | EDC Lancaster County, Center for Regional Analysis *Aug 2026 – Present*

- Automated 25+ federal data series (BLS, BEA, Census, USPS) into a Power BI model with 20 years of history and daily refresh, saving 15+ hours/month.
- Built a Google Apps Script web app with a Gemini AI assistant that flags anomalies, so 6+ non-technical colleagues maintain and explore the data without code.

Data Science Intern | KCC Capital Partners, Sheridan, WY *Jan – May 2026*

- Gathered data on 100+ mining companies with Excel and Seamless.AI, screened it with the firm's built-in AI tools, and reviewed a 100+ buyer shortlist; automated weekly Excel reports, saving 5+ hrs/week.

Data Engineer Intern | KPIM Joint Stock Company, Hanoi, Vietnam *Jan – Apr 2021*

- Built Python and SQL Server ETL pipelines that automated legacy client data workflows, cutting processing time 30%.

SKILLS & CERTIFICATIONS

- **Programming:** Python (pandas, NumPy, scikit-learn, PySpark), SQL (T-SQL, Spark SQL), R, C++, Java
- **ML & Data:** regression, classification, clustering, Bayesian networks, model evaluation; Microsoft Fabric, SQL Server, Power BI, ETL, data modeling, Git
- **Certifications:** Microsoft Fabric Data Engineer Associate (DP-700), Oct 2026; HackerRank SQL (Advanced), Sep 2025